American Computer Science League

2023-2024 ● Contest 1: Short Problems ● Junior Division

1. Computer Number Systems

Convert 2324_{10} to octal.

A. 4442
B. 4244
C. 2444
D. 4424
E. 4224

2. Computer Number Systems

What is the value of $73452_8 - 4261_8 + 752_8$ in base 16?

A. 27771
B. 70143
C. 7063
D. 70163
E. 7043

3. Recursive Functions

Find $f(22)$ given:

$$f(x) = \begin{cases} f(x-4)+1 & \text{if } x \geq 12 \\ 2 \cdot f(x+1) - 3 & \text{if } 8 < x < 12 \\ x - 3 & \text{if } x \leq 8 \end{cases}$$

A. 3
B. 6
C. 9
D. 17
E. 18

4. Recursive Functions

Find $f(f(24))$ given:

$$f(x) = \begin{cases} f\left(\frac{x}{2}\right)+4 & \text{if } x \text{ is even} \\ f(x-1)+2 & \text{if } x \text{ is odd and prime} \\ x & \text{otherwise} \end{cases}$$

Remember that 1 is not a prime number.

A. 21
B. 19
C. 15
D. 11
E. 9

5. What Does This Program Do? (Branching)

In jobs that are paid an hourly wage, overtime is paid at 1.5 times the regular rate for any hours over 40. Which of the following correctly calculates this pay?

a.
```
if hrs > 40 then
     pay = (hrs-40) * rate * 1.5 + hrs * rate
else
     pay = hrs * rate
end if
```

b.
```
pay = hrs * rate
if hrs > 40 then
     pay = pay + (hrs-40) * rate * 0.5
end if
```

c.
```
if hrs > 40 then
     pay = (hrs-40) * rate * 0.5 + hrs * rate
else
     pay = hrs * rate
end if
```

d.
```
pay = 40 * rate
if hrs > 40 then
     pay = pay + 1.5 * (hrs - 40) * rate
end if
```

A. a, b, c
B. a, c, d
C. b, c, d
D. a, c
E. b, c

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1. 计算机数字系统

将 2324_{10} 转换为八进制。

A. 4442
B. 4244
C. 2444
D. 4424
E. 4224

2. 计算机数字系统

$73452_8 - 4261_8 + 752_8$ 的值用十六进制表示为什么？

A. 27771
B. 70143
C. 7063
D. 70163
E. 7043

3. 递归函数

根据下列表达式，求 $f(22)$ ：

$$f(x) = \begin{cases} f(x-4)+1 & \text{如果 } x \geq 12 \\ 2 \cdot f(x+1) - 3 & \text{如果 } 8 < x < 12 \\ x - 3 & \text{如果 } x \leq 8 \end{cases}$$

A. 3
B. 6
C. 9
D. 17
E. 18

4. 递归函数

根据下列表达式，求 $f(f(24))$ ：

$$f(x) = \begin{cases} f\left(\frac{x}{2}\right)+4 & \text{如果 } x \text{ 是偶数} \\ f(x-1)+2 & \text{如果 } x \text{ 既是奇数又是质数} \\ x & \text{其他情况下} \end{cases}$$

注意：1 不是质数。

A. 21
B. 19
C. 15
D. 11
E. 9

5. 这个程序是做什么的？（分支）

对于按时薪结算的工作而言，超过 40 个小时的工作时间，每小时薪资按正常时薪的 1.5 倍计算。下列哪项可以正确计算这类工资？

a.
```
if hrs > 40 then
     pay = (hrs-40) * rate * 1.5 + hrs * rate
else
     pay = hrs * rate
end if
```

b.
```
pay = hrs * rate
if hrs > 40 then
     pay = pay + (hrs-40) * rate * 0.5
end if
```

c.
```
if hrs > 40 then
     pay = (hrs-40) * rate * 0.5 + hrs * rate
else
     pay = hrs * rate
end if
```

d.
```
pay = 40 * rate
if hrs > 40 then
     pay = pay + 1.5 * (hrs - 40) * rate
end if
```

A. a, b, c
B. a, c, d
C. b, c, d
D. a, c
E. b, c